

Differentiation - Set 1

TMUA practice questions by topic. Specification MM6.1-MM6.3, MM8.5. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Find $\frac{dy}{dx}$ when $y = \frac{(3x-2)^2}{x^2}$.
- A $\frac{12}{x^2} + \frac{8}{12x^3}$
- B $\frac{x}{12x^2} - \frac{8}{x^3}$
- C $-\frac{12}{x^2} + \frac{8}{x^3}$
- D $\frac{6(3x-2)}{x^2}$
- E $\frac{12}{x^2}$
- F $9 - \frac{12}{x}$
- 2 Find the coordinates of the stationary points of $y = x^3 - 3x^2 - 9x + 5$ and state their nature.
- A maximum at $(1, -6)$, minimum at $(-3, -22)$
- B minimum at $(-1, 10)$, maximum at $(3, -22)$
- C maximum at $(-1, 10)$, minimum at $(3, -22)$
- D maximum at $(-1, 0)$, minimum at $(3, -32)$
- E maximum at $(-1, 10)$ only
- 3 The tangent to $y = x^2 - 4x + 1$ at the point where $x = 3$ meets the y-axis at
- A $(0, -2)$
- B $(0, 8)$
- C $(0, 4)$
- D $(0, 1)$
- E $(0, -5)$
- F $(0, -8)$

- 4 Find the equation of the normal to the curve $y = \sqrt{x}$ at the point $(4, 2)$.
- A $y = -4x + 16$
B $y = -4x + 18$
C $y = -4x + 2$
D $y = \frac{1}{4}x + 1$
E $y = -\frac{1}{4}x + 3$
F $y = 4x - 14$
- 5 For what range of values of x is $f(x) = 2x^3 - 15x^2 + 24x + 1$ decreasing?
- A $-4 < x < -1$
B $2 < x < 3$
C $x > 1$
D $x < 1$ or $x > 4$
E $x < 4$
F $1 < x < 4$
- 6 The displacement s metres of a particle at time t seconds is $s = t^3 - 6t^2 + 9t$. At what times is the particle momentarily at rest?
- A $t = 2$ only
B $t = 1$ and $t = 3$
C $t = 1$ only
D $t = 3$ only
E $t = 0$ and $t = 3$
F $t = 0, t = 1$ and $t = 3$
- 7 The curve $y = ax^2 + bx$ has a stationary point at $(2, -8)$. Find $a + b$.
- A -6
B -2
C 10
D 2
E -10
F 6

- 8** A rectangle has perimeter 40. What is its maximum possible area?
- A** 80
B 50
C 160
D 400
E 100
F 200
- 9** The curve $y = x^3 + px^2 + qx$ has stationary points at $x = 1$ and $x = 3$. Find q .
- A** 3
B 12
C 9
D -9
E -6
F 6
- 10** The line $y = 4x + c$ is a tangent to $y = x^2 + 2x + 5$. Find c .
- A** 4
B -4
C 1
D 5
E 6
F 3
- 11** Find $\frac{d^2y}{dx^2}$ when $y = \frac{1}{x^2} + 4\sqrt{x}$.
- A** $\frac{3}{x^4} - x^{-\frac{3}{2}}$
B $\frac{3}{x^4} + x^{-\frac{3}{2}}$
C $\frac{3}{x^4} - 2x^{-\frac{3}{2}}$
D $\frac{3}{x^4} - x^{-\frac{3}{2}}$
E $-\frac{3}{2x^4} - x^{-\frac{3}{2}}$
F $\frac{2}{x^3} + 2x^{-\frac{1}{2}}$

- 12** The curve $y = x^2 + kx$ (with $k < 0$) crosses the x -axis at the origin and at one other point P . The gradient of the curve at P is 6. Find k .

- A** 6
- B** -6
- C** -12
- D** -3
- E** 3
- F** -2